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Human foot muscle strength and its association with sprint acceleration, cutting and jumping performance, and kinetics in high-level athletes

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ABSTRACT

The primary objective of this study was to investigate the relationship between metatarsophalangeal joint (MTPj) flexion torque and sprint acceleration, cutting and jumping performance, and kinetics. A secondary aim was to explore this relationship when MTP flexion strength was associated with other foot and lower limb neuromuscular outputs. After an initial MTPj flexion torque assessment using a custom-built dynamometer, 52 high-level athletes performed the following tasks on a force platform system: maximal sprint acceleration, 90-degree cutting, vertical and horizontal jumps, and foot-ankle hops. Their foot posture, foot passive stiffness and foot-ankle reactive strength were assessed using the Foot Posture Index, the Arch Height Index Measurement System and the Foot-Ankle Rebound Jump Test. Ankle plantarflexion and knee extension isometric torque were assessed using an isokinetic dynamometer. During maximal speed sprinting, multiple linear regressions suggested a major contribution of MTPj flexion torque, foot passive stiffness and foot-ankle reactive strength to explain 28% and 35% of the total variance in the effective vertical impulse and contact time. Ankle plantarflexor and quadriceps isometric torques were aggregately contributors of acceleration performance and separate contributors of cutting and jumping performance. In conclusion, MTPj flexion torque was more strongly associated with sprinting performance kinetics especially at high-speed.

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Metatarsophalangeal joint strength; maximum speed; ankle plantarflexors; change of direction; acceleration

1. Introduction

The human foot is a complex structure with multiple joints and degrees of freedom that play an important role in modulating lower limb energetics (Farris et al., 2019; R. E. Smith et al., 2021). Recent biomechanical studies have revealed that approximately 18% of the net centre of mass energy dissipation and 12% of energy generation during rapid tasks such as running or jumping can be attributed to the foot (Farris et al., 2019; Kessler et al., 2020; R. E. Smith et al., 2021). The primary contributors to this energy exchange are the extrinsic foot muscles (EFM), which possess proximal insertions outside the foot, accounting for approximately 13% of energy dissipation and 9% of energy generation (R. E. Smith et al., 2021). Moreover, the foot also acts as an efficient lever during propulsion, facilitating the rapid transmission of force from the ankle plantar flexors to the ground through its intrinsic foot muscles (IFM), which have all their insertions within the foot (Erdemir & Piazza, 2002). Interestingly, studies have shown that when deactivating the IFM during the push-off phase of rapid tasks, both metatarsophalangeal joint (MTPj) stiffness and positive foot-ankle power decreased significantly (Farris et al., 2019; Kessler et al., 2020; R. E. Smith et al., 2021). These findings suggest that the IFM actively stiffen the forefoot during propulsion by generating moments at the MTPj, thereby enhancing lower limb force transmission to the ground (Farris et al., 2020). In addition, the EFM toe flexors (EFMtf), including the *flexor hallucis longus* and *flexor digitorum longus*, also contribute to MTPj function by accelerating the centre of mass during locomotion (Péter et al., 2015). Given the role of ankle plantarflexor muscles in optimizing body acceleration during maximal sprinting (Pandy et al., 2021), cutting (Marshall et al., 2014) and jumping (Vanezis & Lees, 2005), it stands to reason that a "stronger" forefoot would facilitate effective ankle plantarflexor force transmission to the ground during explosive tasks (R. Smith et al., 2022). While the energetic and biomechanical significance of the foot in human locomotion is well established, the literature pertaining to the relationship between MTPj flexion strength and sport performance remains limited. To date, only two studies have investigated this association during jumping and cutting tasks (Yamauchi & Koyama, 2020; Yuasa et al., 2018), revealing moderate correlations ($r = 0.38–0.55$) between "toe gripping strength" with the MTP in neutral position, and

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jump height and vertical impulse in various vertical jumps (Yamauchi & Koyama, 2020), but not with agility tests performance (Yuasa et al., 2018). Conversely, "toe pushing strength" with MTPj in pre-positioned dorsiflexed position was moderately correlated ($r = -0.50$) with cutting performance (Yuasa et al., 2018). The authors also suggested that this experimental setting was better for investigating the strength of human foot muscles since the MTPj undergoes a dorsiflexion moment to absorb energy while power is generated at the midfoot during explosive tasks (Bezodis et al., 2012; McDonald et al., 2016; G. Smith et al., 2014). While these studies represent initial steps to understand the relationship between MTPj flexion strength and sport performance, further investigation is needed, particularly in healthy performance-oriented or even high-level athletes. Consequently, the primary objective of this study was to elucidate the relationships between MTPj maximal isometric flexion torque and the overall performance in maximal sprint acceleration, cutting and jumping tasks. A secondary objective was to explore the relationship between MTPj maximal isometric flexion torque and ground reaction force (GRF) kinetics underlying performance during these tasks. A third objective was to examine how these relationships may vary when considering other foot (foot posture and foot morphological deformation), ankle (maximal ankle plantarflexor isometric torque, foot-ankle reactive strength) and knee variables (maximal knee quadriceps isometric torque). It was hypothesized that MTPj maximal isometric flexion torque variables would be correlated with overall performance, GRF impulse and ratio of forces during jumping and cutting. Furthermore, we hypothesized that incorporating other neuromuscular foot-ankle variables (maximal ankle plantar flexors isometric torque and foot-ankle reactive strength) would enhance the multiple linear regression model for predicting GRF features during jumping and cutting tasks, given the IFM's capacity to modulate the leverage and kinetic output of the ankle plantarflexor torque during propulsion (Farris et al., 2020; R. Smith et al., 2022; R. E. Smith et al., 2021).

2. Methods

2.1. Study design and procedure

This study followed a cross-sectional design with performance and biomechanical data being measured during a single testing session and on the dominant foot only. Limb dominance was determined using three unskilled and skilled tasks (Schneiders et al., 2010). Sprint acceleration, cutting and jumping kinetics were assessed using a segment of ~5.4-m (6×900 mm) force platforms (Kistler, Winterthur, Switzerland) embedded within the laboratory ground. A laptop running BioWare software (version 5.11, Kistler Instruments Inc., Amherst, NY, USA) was used to collect kinetic data for all the explosive tasks, with a sampling rate of 2000 Hz. Standardized shoes with a minimalist index of 95% (Saguaro™) were provided to reduce the effect of different shoe material properties on the performance and kinetic variables measured. All participants gave their written informed consent before enrolment.

2.2. Sample size

Previously mentioned studies showed moderate correlations ($r = 0.38–0.55$) between "toe gripping or pushing strength" and vertical jumping or cutting overall performance (Yamauchi & Koyama, 2020; Yuasa et al., 2018). We evaluated MTPj flexion torque in pre-positioned dorsiflexed position, allowing us to be in the window of optimal angle of torque production (Goldmann & Brüggemann, 2012; Roma et al., 2024; Saeki et al., 2021) while being close to MTPj biomechanical function as they underwent a large dorsiflexion moment during the stance phase of explosive tasks (Deschamps et al., 2020; Goldmann et al., 2013; G. Smith et al., 2014). As MTPj torque could have a greater association with GRF kinetics metrics than with an overall performance metric (Yamauchi & Koyama, 2020), we expected that within our dataset, MTPj maximal isometric flexion torque could reach a greater and larger correlation ($r > 0.7$) than in previous studies. Therefore, our sample size was calculated using the method and equations of Bonnet and Wright to appreciate this desired Pearson correlation coefficient (Bonett & Wright, 2000). Considering a confidence interval width of $\omega = 0.3$, a confidence level of $\alpha = 0.05$ and an estimated $r = 0.70$, a sample of 49 participants was required to obtain a 95% confidence interval for these desired width and coefficient correlation.

2.3. Participants

Fifty-two healthy high-level athletes (5 females; mean ± SD age: 20.1 ± 2.1 years, height: 183.2 ± 10.8 cm, body mass: 78.5 ± 15.5 kg) volunteered to participate in this study. All participants were injury-free and practiced their sport at a national level in soccer ($n = 12$), rugby ($n = 8$), track and field ($n = 14$), basketball ($n = 14$) and handball ($n = 4$). Participants' demographic information sorted by sport is detailed in Table 1.

2.4. Metatarsophalangeal joint (MTPj) maximal isometric flexion torque

MTPj maximal isometric flexion torque in MTPj in a ~ 30° dorsiflexed position was assessed using a custom-built dynamometer with a 6-component force sensor (Nano 25, ATI Industrial Automation, Garner, NC) (Figure 1). This dynamometer shows a good test-retest reliability (ICC = 0.85 [95%CI: 0.73, 0.92]) for measuring maximal isometric MTPj flexion torque in healthy adults (Roma et al., 2024). After a standardized warm-up with submaximal contractions, participants were instructed to "push as hard as they could" during five 5-second contractions with a 1-minute rest period in between (Roma et al., 2024). Force signal was recorded at 1000 Hz using a power-lab data system (16/30-ML880/P, ADInstruments, Bella Vista, Australia), visualized via Labchart software (v7, ADInstrument) and analysed using custom-written Python code. The highest value for the five contractions on the z-axis was considered as the "MTPj maximal isometric pushing force", on the y-axis as the

Table 1. Participants' demographic.

	Overall ($n = 52$)		Soccer ($n = 12$)		Rugby ($n = 8$)		Track & field ($n = 14$)		Basketball ($n = 14$)		Handball ($n = 4$)	
	N Mean	(%) ±SD	N Mean	(%) ±SD	N Mean	(%) ±SD	N Mean	(%) ±SD	N Mean	(%) ±SD	N Mean	(%) ±SD
Male gender	47	(90.4%)	12	(100%)	8	(100%)	9	(64.3%)	14	(100%)	4	(100%)
Age (years)	20.1	±2.1*	19.0	±1.4[c]	20.5	±2.4	21.6	±2.5[ad]	19.4	±1.2[c]	19.8	±0.5
Height (cm)	183.2	±10.8*	179.5	±5.7[d]	188.6	±8.9[c]	172.8	±7.9[bd]	193.6	±7.8[ac]	183.3	±3.4
Body mass (kg)	78.5	±15.5*	74.6	±5.6[bc]	99.8	±18.3[ac]	65.3	±9.6[abd]	83.1	±11.0[c]	77.7	±8.8
BMI (kg/m^2)	23.2	±3.1*	23.2	±1.9[b]	27.9	±3.7[acde]	21.7	±1.9[b]	22.1	±2.4[b]	23.1	±2.6[b]
Weekly sport volume (hour/week)	14.5	±5.7*	19.9	±0.3[bcde]	15.6	±4.8[ac]	9.2	±2.9[abd]	15.5	±6.0[ac]	11.1	±5.3[a]
Lower limb dominance												
Right	47	(90.4%)	10	(80%)	8	(100%)	14	(100%)	13	(97.7%)	2	(50%)
Left	5	(9.6%)	2	(20%)	0	(0%)	0	(0%)	1	(2.3%)	2	(50%)
CAIT score (dominant foot)	25.7	±4.4*	28.1	±3.2[bed]	24.8	±3.7[a]	25.8	±5.3	24.9	±3.9[a]	23.0	±5.2[a]

BMI, Body Mass Index; CAIT, Cumberland Ankle Instability Tool.
difference between-sports following a one-way analysis of variance (ANOVA) with different group sample size.
a: difference with soccer ($p < 0.05$).
b: difference with rugby ($p < 0.05$).
c: difference with track & field ($p < 0.05$).
d: difference with basketball ($p < 0.05$).
e: difference with handball ($p < 0.05$).

Figure 1. Metatarsophalangeal joint maximal isometric flexion torque measurement using a custom-built dynamometer with a 6-component force sensor.

"MTPj maximal isometric gripping force" and on the yz-resultant time series as the "MTPj maximal isometric total force". Associated torques around the MTPj were then assessed by multiplying the absolute force by the length of the first ray (subtraction of the total foot length by the truncated foot length) using the Arch Height Index Measurement System (JAKTOOL Corporation, Cranberry, NJ) (Butler et al., 2008).

2.5. Foot posture and foot morphological deformation

Foot posture was assessed in a relaxed bipedal standing position using the Foot Posture Index–6 item version (FPI-6) (Redmond et al., 2006) while the foot morphological deformation was assessed using the Arch Height Index Measurement System (Butler et al., 2008). Following previous methodology, foot 1-dimensional morphological deformation was then assessed using "arch height flexibility" while the foot 2 and 3-dimensional morphological deformations were assessed using "foot mobility magnitude" and "quarter-ellipsoid foot" (Tourillon et al., 2023).

2.6. Maximal isometric quadriceps (MIQt) and ankle plantarflexor (MIPFt) torque

Both torques were measured on an isokinetic dynamometer (Con-Trex MJ, CMV AG, Dubendorf, Switzerland), with the MIQt assessed in a seated upright position (90° knee angle), and the MIPFT assessed in a prone position with the ankle in a neutral position. The data collection and analysis followed the same procedure as for MTPj maximal isometric flexion torque. The highest torque value among all five contractions was retained for statistical analysis.

2.7. Sprint acceleration, cutting and jumping tasks

The detailed study protocol for these tasks is presented in supplementary material 1. For the sprint acceleration task, participants performed maximal accelerations in three different conditions (Schache et al., 2019): "high-acceleration", "medium-acceleration", and "low-acceleration", which allowed us to analyse the GRF over the 0 to 6, 7 to 13 and 30 to 36 m sections, respectively (Schache et al., 2019). For the cutting task, participants were instructed to "run as fast as possible" for 5 m, make a single and complete foot contact with the force platform and then perform a 90° turn before running "as fast as possible" again for 5 m to the finish line. The jumping tasks consisted in a "vertical and horizontal CMJ" where participants were instructed to "jump as fast and as high as possible" or "as fast and as far as possible" while constantly keeping their hands on their hips. For all these explosive tasks, the trial showing the best performance was retained for statistical analysis. Performance was assessed as the best 10-m sprint and cutting time using the timing gates system (Microgate, Bolzano, 113 Italy), jump height for "vertical CMJ" and horizontal peak power for "horizontal CMJ" based on the impulse-momentum relationship (Linthorne, 2001). GRF signals were filtered using a Butterworth low-pass digital filter at a cut-off frequency of 50 Hz. Using a custom MATLAB script (MathWorks, R2022b, Natick, MA, USA), discrete GRF variables were calculated

similarly to previous sprint (Nagahara et al., 2018; Schache et al., 2019), cutting (Havens & Sigward, 2015; Spiteri et al., 2013) and jumping (Linthorne, 2001; McMahon et al., 2018) biomechanics studies (see supplementary material 1, 2, 3 and 4). For each explosive task, these variables included (a) the impulse (integral over time) of the vertical (Fz) and/or horizontal (Fy) and/or medio-lateral GRF (Fx); (b) the effective impulse of the vertical GRF i.e., the product of the stance phase duration by the average vertical GRF applied above body-weight; (c) the net impulse of the horizontal GRF and the impulse of each component of the horizontal GRF, i.e., the negative (braking) and positive (propulsive) components; (d) the ratio of forces (RF), i.e the ratio of the step- or phase averaged Fy or Fx component divided by the resultant of the step- or phase averaged GRF (FTot); and (e) the contact time defined by the events of foot-strike and toe-off from the raw GRF data (Fz threshold of 20 N for sprinting and 10 N for cutting). Finally, participants performed a jumping task called "Foot-Ankle Rebound Jumps" (FARJ), which was used to evaluate foot-ankle reactive strength and stretch-shortening cycle (SSC) capacities (Bosco et al., 1982; Nagahara et al., 2014). Participants were instructed to jump "as high as possible" while keeping their lower limbs fully straight, and to push against the ground "as quickly as possible" with only a plantarflexion of the ankle and the MTPj during eight jumps (Bosco et al., 1982). From the contact and flight time measured with an optoelectronic system (Optojump Next, Microgate, Bolzano, 113 Italy) the mean reactive strength index of four jumps (excluding the first and last two jumps of the series) was calculated as the ratio of jump height (using the flight time method) to contact time (Nagahara et al., 2014).

2.8. Statistical analyses

Descriptive statistics for all variables were determined for the entire group using mean and standard deviations. To assess the relationship between relative MTPj maximal isometric flexion torque variables (pushing, gripping and total) and the overall performance during sprint acceleration, cutting and jumping tasks (1st objective of the study) a Pearson correlation coefficient analysis was used. To assess the relationship between the same MTPj maximal isometric flexion torque variables and GRF kinetics underlying performance in each explosive task (2nd objective of this study), the same procedure was applied. Thresholds for the interpretation of small ($0.1 < r < 0.3$), moderate ($0.3 < r < 0.5$), large ($0.5 < r < 0.7$) and very large ($r > 0.7$) correlations using the values proposed by Hopkins et al. (2009) were applied. Finally, to answer our third research question, a stepwise backward multiple linear regression was used, adding other foot variables (posture, morphological deformation) and lower-limb neuromuscular variables (MIQt, MIPFt and foot-ankle reactive strength) in combination with the MTPj maximal isometric flexion torque variables. The level of significance was set at $p < 0.05$ and all data were analysed using JASP (Amsterdam 0.12.2.0).

3. Results

Descriptive statistics and correlation coefficients between the independent variables (MTPj flexion strength) and all the dependent variables (foot posture, foot morphological deformation, lower-limb neuromuscular capacities and overall performance during each explosive task) are shown in Table 2. Descriptive statistics of the GRF variables during sprint acceleration, cutting, and jumping are presented in supplementary material 2, 3 and 4. Regarding the first aim of this study, relative MTPj maximal isometric torque variables were moderately associated ($r = -0.33$, $p = 0.02$) with the 10-m sprint time during high-acceleration condition (i.e., improved performance) for pushing and total torque, and with horizontal jumping performance ($r = -0.35$, $p = 0.01$) (i.e., lower performance) for gripping torque (Table 2). Regarding the second aim, relative MTPj total and pushing torques were moderately associated with relative vertical impulse ($r = 0.42$, $p = 0.03$) and contact time ($r = -0.41$, $p = 0.04$) (i.e., improved performance) during maximal speed sprinting. Regarding the last aim of this study, the stepwise backward multiple regression analysis showed that relative MTPj total and pushing torques are the greatest contributors to explain relative effective vertical impulse (adjusted $r^2 = 0.28$, $p = 0.008$) and contact time (adjusted $r^2 = 0.35$, $p = 0.003$) in addition to foot morphological deformation and foot-ankle reactive strength, respectively (Figure 3). Additionally, when aggregated, relative MIPFt and MIQt were the greatest contributors to explain acceleration performance (adjusted $r^2 = 0.27$–0.36, $p < 0.001$) (Figure 2) while variance in cutting performance was explained by relative MIPFt, foot-ankle reactive strength and foot posture (adjusted $r^2 = 0.35$, $p < 0.001$) (Figure 4), and jumping performance was explained by relative MIQt, foot morphological deformation and relative MTPj gripping torque (adjusted $r^2 = 0.26$–0.38, $p < 0.001$) (Figure 5).

4. Discussion

The primary objective of this study was to investigate the relationship between relative MTPj isometric flexion torque and sprint acceleration, and cutting and jumping performance and kinetics. A secondary objective was to explore the same relationship when adding other foot and neuromuscular lower limb parameters.

4.1. Acceleration

During the early (0 to 10 m) and intermediate (8 to 18 m) phases of the acceleration, we observed that the association of relative MIPFt and relative MIQt explained 27% and 36% of the total variance in 10-m sprint time respectively. This heightened contribution of the relative MIPFt and MIQt on acceleration performance, as opposed to the maximal speed phase, could be attributed to the distinct biomechanics of these joints during those specific phases. In steady-state sprinting, the joint power profile at the ankle and the knee shows a sinusoidal pattern with a period of power absorption during early stance followed by power generation during late stance (Schache et al., 2015). Conversely, during maximal acceleration, the ankle and knee joints show a bias towards power generation and positive work, although a small amount of power absorption and negative work still occurs at the ankle during early stance even in the high-acceleration condition (Schache et al., 2019). These joint mechanical profiles suggest that the ankle

Table 2. Descriptive statistics and correlation coefficients between MTPj flexion torque variables and foot posture, foot morphological deformation, lower-limb neuromuscular capacities, sprinting acceleration, cutting and jumping performance.

	Overall ($n = 52$)		Relative pushing torque (Nm/kg)		Relative gripping torque (Nm/kg)		Relative total torque (Nm/kg)	
	Mean	*±SD*	*r [95% CCI]*	*p-value*	*r [95% CCI]*	*p-value*	*r [95% CCI]*	*p-value*
MTPj maximal isometric flexion strength								
Absolute pushing absolute force (N)	321.9	±80.3	**0.84 [0.73, 0.90]**	**<0.001****	**0.37 [0.10, 0.59]**	**0.008****	**0.83 [0.72, 0.90]**	**<0.001****
Relative pushing torque (Nm/kg)	0.30	±0.07			**0.43 [0.18, 0.64]**	**0.002****	**0.99 [0.99, 0.99]**	**<0.001****
Absolute gripping force (N)	88.0	±34.2	**0.32 [0.05, 0.55]**	**0.02***	**0.92 [0.86, 0.95]**	**<0.001****	**0.36 [0.10, 0.58]**	**0.01***
Relative gripping torque (Nm/kg)	0.08	±0.03	**0.43 [0.18, 0.64]**	**0.002****			**0.48 [0.23, 0.67]**	**<0.001****
Absolute total force (N)	327.5	±80.9	**0.83 [0.73, 0.90]**	**<0.001****	**0.41 [0.15, 0.62]**	**0.003****	**0.83 [0.72, 0.90]**	**<0.001****
Relative total torque (Nm/kg)	0.31	±0.08	**0.99 [0.99, 0.99]**	**<0.001****	**0.48 [0.23, 0.67]**	**<0.001****		
Foot posture								
FPI score	3.8	±3.0	0.26 [−0.01, 0.50]	0.06	−0.17 [−0.43, 0.12]	0.24	0.25 [−0.02, 0.49]	0.07
Foot morphological deformation								
Arch height flexibility (mm/kN)	7.8	±3.5§	−0.16 [−0.41, 0.12]	0.25	0.02 [−0.26, 0.30]	0.90	−0.15 [−0.40, 0.13]	0.30
Foot mobility magnitude (cm)	0.71	±0.23	−0.18 [−0.43, 0.09]	0.19	−0.16 [−0.42, 0.12]	0.26	−0.19 [−0.44, 0.09]	0.19
Quarter-ellipsoid foot								
Δ Surface area (cm^2)	5.5	±5.3	0.12 [−0.16, 0.38]	0.39	−0.23 [−0.48, 0.05]	0.11	0.10 [−0.18, 0.36]	0.49
Δ Volume (cm^3)	−2.6	±27.9§	0.21 [−0.07, 0.46]	0.13	−0.16 [−0.42, 0.12]	0.26	0.19 [−0.09, 0.44]	0.19
Δ Surface area to volume ratio (%)	0.67	±0.50§	−0.27 [−0.50, 0.01]	0.06	0.03 [−0.25, 0.30]	0.86	−0.24 [−0.48, 0.03]	0.08
Foot-ankle reactive strength								
Reactive strength index (m/s)	0.47	±0.25§	0.23 [−0.05, 0.47]	0.10	0.12 [−0.17, 0.38]	0.42	0.23 [−0.04, 0.48]	0.10
Maximal isometric quadriceps torque								
Absolute torque (Nm)	268.0	±77.6§	0.05 [−0.23, 0.32]	0.74	0.16 [−0.13, 0.42]	0.28	0.05 [−0.23, 0.31]	0.75
Relative torque (Nm/kg)	3.4	±0.7	**0.32 [0.05, 0.55]**	**0.02***	0.28 [−0.00, 0.52]	0.05	**0.32 [0.05, 0.54]**	**0.02***
Maximal isometric plantarflexor torque								
Absolute torque (Nm)	130.4	±36.7§	0.02 [−0.26, 0.29]	0.92	0.01 [−0.27, 0.29]	0.96	0.00 [−0.27, 0.28]	0.97
Relative torque (Nm/kg)	1.7	±0.5	0.27 [−0.00, 0.51]	0.05	0.08 [−0.20, 0.35]	0.57	0.26 [−0.02, 0.50]	0.07
High-acceleration sprinting condition								
10-m sprint time (s)	1.90	±0.11	**-0.33 [−0.55, −0.06]**	**0.02***	−0.07 [−0.34, 0.21]	0.62	**-0.32 [−0.54, −0.05]**	**0.02***
Medium-acceleration sprinting condition								
10-m sprint time (s)	1.33	±0.06	−0.26 [−0.50, 0.01]	0.06	−0.17 [−0.43, 0.12]	0.25	−0.25 [−0.49, 0.03]	0.08
Low-acceleration sprinting condition								
10-m sprint time (s)	1.18	±0.07	−0.10 [−0.47, 0.30]	0.61	−0.08[−0.45, 0.32]	0.71	−0.09 [−0.46, 0.31]	0.66
Cutting								
COD time (s)	0.94	±0.09	−0.25 [−0.49, 0.03]	0.08	0.13 [−0.15, 0.39]	0.36	−0.24 [−0.48, 0.04]	0.09
Vertical countermovement jump								
Jump height (cm)	38.39	±5.40	0.03 [−0.25, 0.30]	0.85	−0.13 [−0.40, 0.15]	0.37	0.00 [−0.27, 0.28]	0.97
Horizontal countermovement jump								
Relative horizontal peak power (W/kg)	19.76	±3.89§	−0.08 [−0.34, 0.20]	0.60	**-0.35 [−0.58, −0.08]**	**0.01***	−0.10 [−0.37, 0.18]	0.47

MTPj, Metatarsophalangeal joint; FPI, Foot Posture Index; Δ, Delta; COD, Change of direction.
§difference between-sports (track & field, rugby, soccer, basketball, handball) following a one-way analysis of variance (ANOVA) with different group sample size.
*$p < 0.05$.
**$p < 0.01$.

Model 1	Adj. R^2	RMSE	p-value
High-acceleration condition: 10-m sprint time	0.27	0.091	<0.001
Independent variables	**Coefficient**	**t**	**p-value**
Relative maximal isometric quadriceps torque	-0.062	-3.399	0.001
Relative maximal isometric plantarflexor torque	-0.088	-2.250	0.029
Constant	2.260	27.972	<0.001

Model 2	Adj. R^2	RMSE	p-value
Medium-acceleration condition: 10-m sprint time	0.36	0.048	<0.001
Independent variables	**Coefficient**	**t**	**p-value**
Relative maximal isometric quadriceps torque	-0.043	-4.431	<0.001
Relative maximal isometric plantarflexor torque	-0.045	-2.143	0.037
Constant	1.551	36.190	<0.001

RMSE: Root Mean Square Error; Adj.: adjusted

Figure 2. Multiple linear regression analysis for sprint performance (in s) during the high and medium-acceleration conditions.

Model 3	Adj. R^2	RMSE	p-value
Low-acceleration condition: Relative effective vertical impulse	0.28	0.013	0.008
Independent variables	**Coefficient**	**t**	**p-value**
Relative MTPj maximal isometric total torque	0.069	1.885	0.072
Foot mobility magnitude	-0.024	-2.380	0.026
Constant	0.110	7.603	<0.001

Model 4	Adj. R^2	RMSE	p-value
Low-acceleration condition: Contact time	0.35	0.010	0.003
Independent variables	**Coefficient**	**t**	**p-value**
Relative MTPj maximal isometric pushing torque	-0.057	-1.966	0.062
Foot-ankle reactive strength index	-0.027	-3.037	0.006
Constant	0.149	16.599	<0.001

MTPj: Metatarsophalangeal Joint; RMSE: Root Mean Square Error; Adj.: adjusted

Figure 3. Multiple linear regression analysis for relative effective vertical impulse (in BW.S) and contact time (in s) during the low-acceleration condition.

Model 5	Adj. R^2	RMSE	p-value
Cutting: COD time	0.35	0.069	<0.001
Independent variables	**Coefficient**	**t**	**p-value**
Relative maximal isometric plantarflexor torque	-0.054	-1.769	0.083
Foot-ankle reactive strength index	-0.150	-3.708	<0.001
Foot posture score	-0.011	-3.273	0.002
Constant	1.145	21.906	<0.001

RMSE: Root Mean Square Error; Adj.: adjusted ; COD: Change of Direction

Figure 4. Multiple linear regression analysis for change of direction performance (in s) during cutting.

Model 6	Adj. R^2	RMSE	p-value
Vertical CMJ: Jump height	0.23	0.048	<0.001
Independent variables	**Coefficient**	**t**	**p-value**
Relative maximal isometric quadriceps torque	0.027	2.766	0.008
Quarter-ellipsoid foot Δ surface area	0.003	2.132	0.038
Constant	0.276	8.475	<0.001
Model 7	**Adj. R^2**	**RMSE**	**p-value**
Vertical CMJ: Relative peak power	0.26	5.622	<0.001
Independent variables	**Coefficient**	**t**	**p-value**
Relative maximal isometric quadriceps torque	3.257	2.808	0.007
Quarter-ellipsoid foot Δ surface area	0.073	2.456	0.018
Constant	45.079	11.127	<0.001
Model 8	**Adj. R^2**	**RMSE**	**p-value**
Horizontal CMJ: Relative horizontal peak power	0.38	3.060	<0.001
Independent variables	**Coefficient**	**t**	**p-value**
Relative maximal isometric quadriceps torque	2.946	4.637	<0.001
Relative MTPj maximal isometric gripping torque	-61.710	-4.299	<0.001
Constant	14.784	6.707	<0.001

MTPj: Metatarsophalangeal Joint; RMSE: Root Mean Square Error; Adj.: adjusted; Δ: delta ; CMJ: Countermovement jump

Figure 5. Multiple linear regression analysis for jump performance (in cm) and relative peak power (in W/kg) during vertical countermovement jump and relative horizontal peak power (in W/kg) during horizontal countermovement jump.

joint shows the strongest positive linear relationship with forward acceleration magnitude followed by the hip and the knee joints, highlighting their roles in acceleration performance (Schache et al., 2019). In a separate study, quadriceps muscles (*vastii* and *rectus femoris*) and the ankle plantarflexors demonstrated the highest potential for generating support force by accelerating the body upward throughout the stance (Pandy et al., 2021). While the quadriceps primarily contributes to vertical force production orientation during acceleration, it also serves as a brake (potentially decelerating running speed) (Pandy et al., 2021). Thus, in the horizontal direction, ankle plantarflexors and especially the *gastrocnemii* play a pivotal role in accelerating the body forward by generating propulsive impulse across all foot contacts during sprint acceleration (Pandy et al., 2021). Our findings indicate that the athletes with the superior acceleration tend to exhibit greater relative MIPFt and MIQt, enabling them to produce high amounts of force in both vertical and horizontal directions, despite the absence of significant relationship with GRF features in our study. These outcomes align with previous studies indicating that relative isometric or isokinetic quadriceps torque was associated with 5, 10 and 15-m sprint time in athletes from various sports ($r = -0.44$ to -0.71) (Boraczyński et al., 2020; Dowson et al., 1998; Newman et al., 2004), albeit not with the personal best 100-m sprint time in sprinters (Hori et al., 2021; Miller et al., 2021). Relative isometric quadriceps torque appears to be a crucial contributor of sprint performance over short sprint distances (10 and 20 m) but not over longer sprint distances (i.e., in a high-speed, low-acceleration context) in trained athletes from various acceleration-based sports. Regarding relative ankle plantarflexor isometric torque, this study is the first to demonstrate its contribution to short sprint distance performance since only one study highlighted that absolute but not relative ankle plantarflexor isokinetic torque was significantly correlated ($r = -0.51$) with 15-m sprint time (Dowson et al., 1998).

4.2. Maximal speed

During the sprinting acceleration task at maximum speed (~30 to 36 m from the start line), we observed that 28% and 37% of the variance in relative effective vertical impulse and contact time could be accounted for by relative MTPj maximal isometric pushing torque, foot morphological deformation and foot-ankle reactive strength. These results hold significant implications, given that previous studies highlight the importance of a large vertical force production over a short contact time during the maximal speed phase for achieving a better sprint performance (Nagahara et al., 2018). The upright body posture adopted during maximal running speed likely increases the reliance on the ankle joint to transmit a large amount of force to the ground (Nagahara et al., 2014), as the ankle joint showed the greatest joint power compared to the knee and the hip joints during the support phase of maximal speed sprinting (Schache et al., 2019). However, it has also been shown that

ankle plantarflexors (*soleus* and *gastrocnemius*) primarily influence net propulsive and vertical impulses throughout the entire acceleration phase (Pandy et al., 2021); our study revealed that relative MIPFt was not the predominant contributor in producing effective vertical impulse within a short duration at high speed. This discrepancy could be attributed to the ankle joint progressively transitioning towards elastic strain energy return as running speed approaches maximal limits (Schache et al., 2019). This spring-like feature is not solely dependent on the Achilles tendon but also on the foot's medial longitudinal arch. The foot's medial arch lowers and compresses to store energy in its passive (plantar ligaments and aponeurosis) and active (IFM and EFM) components before returning positive mechanical energy whilst energy is also simultaneously absorbed at the MTPj (Bezodis et al., 2012; McDonald et al., 2016). These SSC capacities around the foot-ankle complex could be assessed using the FARJ, as it primarily engages the MTP and ankle joints (Bosco et al., 1982; Nagahara et al., 2014). In this study, the reactive strength index during this test was significantly associated with acceleration during the later phase of sprint acceleration ($r = 0.48$–0.54), underscoring the importance of improving dynamic leg stiffness in order to achieve higher maximal sprint speed (Nagahara et al., 2014). The latter study and the present results suggest that possessing greater forefoot strength and foot passive stiffness is crucial for effectively transmitting vertical force, while a combination of a higher forefoot strength and foot-ankle reactive strength enables this transmission within a short time. This may elucidate why sprinters demonstrate greater foot-ankle joint stiffness (Stefanyshyn & Nigg, 1998) and thicker foot muscles than runners (Tanaka et al., 2019). Therefore, despite its relatively small size, the MTPj deserves consideration in sprint training, as evidenced by a previous study demonstrating improved 50-m sprint time performance following an 8-week toe flexion strength protocol in healthy active males (Hashimoto & Sakuraba, 2014).

4.3. Cutting

During the "90° anticipated" cutting task, we discovered that increased relative MIPFt, foot-ankle reactive strength and a more supinated foot posture could collectively account for 35% of the variance in COD performance. This finding aligns with previous research demonstrating a strong correlation between peak concentric ankle power and plantarflexion moment, and the time to complete a 75° cutting task ($r = -0.65$ to -0.77) (Marshall et al., 2014). COD speed encompasses two primary components: technical skills including accelerating, top speed, decelerating and turning as well as physical skills including speed, strength, power and reactive strength (Turner et al., 2022). Therefore, while multiple muscle groups contribute to the multijoint action of cutting, our results highlight the significance of ankle complex strength and reactive strength in cutting performance. This may be attributed to the ankle's predominant role in negative and positive work production during anticipated 90° cutting compared to hip and knee joints (David et al., 2018). The ankle plantarflexors especially serve as the primary propellers during the late stance of a 45° cutting task and play a crucial role in vertical support alongside the quadriceps, in particular the *vastii* (Maniar et al., 2019). However, we did not observe any significant relationships between relative MIQt and cutting time or GRF variables, suggesting that athletes in our study may have employed a strategy with a greater reliance on the ankle complex during the 90° cutting task. Regarding the foot joint complex, our results showed that a more neutral and supinated static foot posture may be associated with better ankle plantarflexors force transmission and foot-ankle SSC function. Nevertheless, these results contrast with our hypothesis and a prior research showing that relative MTPj pushing strength was moderately correlated with agility tests performance ($r = -0.50$) (Yuasa et al., 2018). This disparity may be explained by the fact that the foot performed approximately 14% of the negative work and only 3% of positive work during a 45° cutting task (Liew et al., 2020). Consequently, although the MTPj serves as the only ground contact point for most athletes in our study, IFM and EFMtf torque were not associated with cutting performance or GRF features. Therefore, it is conceivable that other EFM such as invertors, evertors or dorsiflexors may play a more substantial role in GRF transmission function in comparison to IFM and EFMtf, especially considering the amount of mediolateral forces encountered during cutting (Wu et al., 2022).

4.4. Jumping

In the jumping tasks, we discovered that increased relative MIQt and higher foot flexibility ("quarter foot ellipsoid variation Δ surface area") accounted for 23 and 26% of the variance in jump height and relative peak power during vertical CMJ respectively. These findings are consistent with a significant body of literature indicating a relationship between relative maximal quadriceps isometric torque and vertical CMJ performance (Boraczyński et al., 2020; Konrad et al., 2021; Križaj et al., 2019; Wdowski et al., 2023). This association ranged from moderate ($r = 0.33$) (Konrad et al., 2021) to very large ($r = 0.70$) (Boraczyński et al., 2020) in isometric mode and large ($r = 0.55$ to 0.66) (Križaj et al., 2019; Wdowski et al., 2023) in isokinetic mode. Although the exact biomechanical contribution of the knee joint during vertical CMJ differs across studies, it consistently plays a substantial role (ranging from 33% to 49%) and serves as a determinant joint for vertical CMJ performance (Kotsifaki et al., 2021; Vanezis & Lees, 2005). An unexpected discovery concerning the foot-ankle complex was the contribution of foot passive flexibility to jump height and relative peak power during vertical CMJ. This finding contrasts with prior studies suggesting that foot arch height, foot posture, or foot morphological deformations are not associated with vertical jump performance or ground reaction force (GRF) variables (Tourillon et al., 2023; Yamauchi & Koyama, 2020). While previous authors have debated the clinical relevance of morphological parameters in relation to vertical jumping (Tourillon et al., 2023) and have shown only moderate associations between toe flexor strength (MTPj gripping strength) and CMJ height (Yamauchi & Koyama, 2020), none of the MTPj maximal isometric flexion torque variables in our study were associated with CMJ performance or kinetic variables, contrary to our hypothesis. These differences raise questions about the

potential role of toe flexor strength in vertical jumping performance. While several intervention studies on jumping have reported increased toe flexion strength after various training programmes (Goldmann et al., 2013; Hashimoto & Sakuraba, 2014; Kokkonen et al., 1988; Unger & Wooden, 2000), these gains did not consistently translate into improved vertical jump performance (Goldmann et al., 2013; Hashimoto & Sakuraba, 2014). That being said, all studies showed an improvement in the horizontal jump performance (Goldmann et al., 2013; Hashimoto & Sakuraba, 2014; Kokkonen et al., 1988; Unger & Wooden, 2000), suggesting a potentially greater and systematic effect of toe flexor strength on horizontally-oriented jumping tasks than vertically-oriented ones. However, the last finding of this study suggested that the association of relative MIQt and relative MTPj maximal isometric gripping torque explained 38% of the variance in horizontal peak power during horizontal CMJ. Despite the fact that horizontal jump length mainly results from the hip and ankle joint with a knee contribution limited to 13% (Kotsifaki et al., 2021), this result suggests that a higher MIQt and a lower MTPj gripping maximal isometric relative torque are important for horizontal CMJ performance. Overall, our results suggest that the force coming from gripping movements of the toes during a MTPj isometric flexion could be detrimental while the force coming from the pushing movements is not. Given that this study is the first, to our knowledge, to raise this point, further biomechanical and interventional studies are necessary to elucidate the influence of MTPj flexion strength on both vertical and horizontal jumping performance.

4.5. Training implications

Considering the biomechanical variables associated with performance in sprint acceleration, cutting, and jumping tasks, we suggest that distinct joints and neuromuscular capacities may prove particularly valuable in enhancing sports physical performance outcomes. Consequently, we advocate for the incorporation of forefoot maximal strength and foot-ankle reactive strength enhancement into athletic development programmes, especially when targeting improvements in maximal speed performance. This may be achieved using exercises involving maximal MTPj isometric flexion in dorsiflexed position with forward lean body overloading position (Goldmann et al., 2013) and fast SSC foot-ankle plyometric exercises such as foot-ankle rebound jumps, pogo hops or drop-jumps (with athletes instructed to execute small countermovement amplitudes) (Griffin et al., 2021; Nagahara et al., 2014). Furthermore, we advocate for a heightened emphasis on developing ankle plantarflexor maximal strength and foot-ankle reactive strength to enhance cutting and acceleration performance. Furthermore, we advocate for a heightened emphasis on developing ankle plantarflexor maximal strength and foot-ankle reactive strength to enhance cutting and acceleration performance. Exercises targeting ankle plantarflexor maximal strength should encompass both the *gastrocnemii*, through calf-raises in a knee-extended position, and the *soleus*, through calf-raises in a knee-flexed position. This can be facilitated using equipment such as a smith machine or a guided barbell. Lastly, in alignment with previous research, our findings underline the ongoing relevance of developing quadriceps maximal strength, given its pivotal role in acceleration and jumping performance.

4.6. Limitations

Firstly, experimental data for the different sprint acceleration conditions were recorded across various trials with three distinct starting points. This approach, previously used (Schache et al., 2019), was required by the dimensions of the force plate system (see methods). However, descriptive statistics of experimental GRF data during sprinting acceleration are presented in supplementary material 2 and closely align with data from single trials on a ~52 m force platform segment (Nagahara et al., 2018). Secondly, we only performed maximal isometric contractions at one specific angle for each joint (MTPj, knee and ankle) because of experimental time constraints. Thus, some findings could be the consequence of a lack of specificity between the isometric strength measurements and the dynamic nature of each explosive task studied. Thirdly, we did not evaluate the hip joint flexor's or extensor's strength because of experimental time constraints. Due to its contribution to explosive tasks, as previously discussed, it may be possible that adding hip joint maximal strength to our models could have explained explosive tasks GRF performance kinetics to a greater extent. Fourthly, data were gathered from a sample of young (≤28 years) high-level athletes with a greater proportion of males (90%). Whilst our cohort represents a sample of high-level athletes practicing their sport at a national level, it is plausible that the relationships between all the studied variables could vary in less skilled, older athletes, or in a more gender-balanced cohort. Finally, our sample size was based on an expected high correlation of $r = 0.7$ which led to a convenient smaller sample size (Bonett & Wright, 2000) and a potential underpower of the evaluated sample, as no correlation in this study reached that desired value.

5. Conclusion

In conclusion, this study shed light on the relationship between forefoot strength and sprint performance in a cohort of high-level athletes for the first time. Particularly, the experiment unveiled that maximal isometric pushing and total torque at the metatarsophalangeal joint (MTPj) were the primary determinants of performance kinetics during maximal speed sprinting, while their significance was less pronounced in other explosive tasks. This underscores the notion that forefoot strength capacity may exert a more substantial impact on horizontally-oriented explosive movements at high velocity than on vertically or medio-laterally-oriented ones.

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Protocol registration

The research presented in this study was conducted at the Interuniversity Laboratory of Human Movement Sciences and the study protocol was a priori approved by the local ethical committee (IRBN1162021/CHUSTE).

References

Bezodis, N. E., Salo, A. I., & Trewartha, G. (2012, May). Modeling the stance leg in two-dimensional analyses of sprinting: Inclusion of the MTP joint affects joint kinetics. *Journal of Applied Biomechanics*, *28*(2), 222–227. https://doi.org/10.1123/jab.28.2.222

Bonett, D. G., & Wright, T. A. (2000, March 1). Sample size requirements for estimating Pearson, Kendall and Spearman correlations. *Psychometrika*, *65*(1), 23–28. https://doi.org/10.1007/BF02294183

Boraczyński, M., Boraczyński, T., Podstawski, R., Wójcik, Z., & Gronek, P. (2020, October). Relationships between measures of functional and isometric lower body strength, aerobic capacity, anaerobic power, sprint and countermovement jump performance in professional soccer players. *Journal of Human Kinetics*, *75*(1), 161–175. https://doi.org/10.2478/hukin-2020-0045

Bosco, C., Tarkka, I., & Komi, P. (1982, June). Effect of elastic energy and myoelectrical potentiation of triceps surae during stretch-shortening cycle exercise. *International Journal of Sports Medicine*, *03*(3), 137–140. https://doi.org/10.1055/s-2008-1026076

Butler, R. J., Hillstrom, H., Song, J., Richards, C. J., & Davis, I. S. (2008, March 1). Arch height index measurement system. *Journal of the American Podiatric Medical Association*, *98*(2), 102–106. https://doi.org/10.7547/0980102

David, S., Mundt, M., Komnik, I., & Potthast, W. (2018, December 1). Understanding cutting maneuvers – the mechanical consequence of preparatory strategies and foot strike pattern. *Human Movement Science*, *62*, 202–210. https://doi.org/10.1016/j.humov.2018.10.005

Deschamps, K., Matricali, G., Peters, H., Eerdekens, M., Wuite, S., Leardini, A., & Staes, F. (2020, July 26). Contribution of foot joints in the energetics of human running. *Computer Methods in Biomechanics and Biomedical Engineering*, *23*(10), 557–563. https://doi.org/10.1080/10255842.2020.1746287

Dowson, M. N., Nevill, M. E., Lakomy, H. K., Nevill, A. M., & Hazeldine, R. J. (1998, April). Modelling the relationship between isokinetic muscle strength and sprint running performance. *Journal of Sports Sciences*, *16*(3), 257–265. https://doi.org/10.1080/026404198366786

Erdemir, A., & Piazza, S. J. (2002, June). Rotational foot placement specifies the lever arm of the ground reaction force during the push-off phase of walking initiation. *Gait & Posture*, *15*(3), 212–219. https://doi.org/10.1016/S0966-6362(01)00192-8

Farris, D. J., Birch, J., & Kelly, L. (2020, July). Foot stiffening during the push-off phase of human walking is linked to active muscle contraction, and not the windlass mechanism. *Journal of the Royal Society, Interface / the Royal Society*, *17*(168), 20200208. https://doi.org/10.1098/rsif.2020.0208

Farris, D. J., Kelly, L. A., Cresswell, A. G., & Lichtwark, G. A. (2019). The functional importance of human foot muscles for bipedal locomotion. *Proceedings of the National Academy of Sciences*, *116*(5), 1645–1650. https://doi.org/10.1073/pnas.1812820116

Goldmann, J. P., & Brüggemann, G. P. (2012, August). The potential of human toe flexor muscles to produce force: Toe flexor muscle strength. *Journal of Anatomy*, *221*(2), 187–194. https://doi.org/10.1111/j.1469-7580.2012.01524.x

Goldmann, J. P., Sanno, M., Willwacher, S., Heinrich, K., & Brüggemann, G. P. (2013). The potential of toe flexor muscles to enhance performance. *Journal of Sports Sciences*, *31*(4), 424–433. https://doi.org/10.1080/02640414.2012.736627

Griffin, C., Daniels, K., Hill, C., Franklyn-Miller, A., & Morin, J. B. (2021, December). A criteria-based rehabilitation program for chronic mid-portion achilles tendinopathy: Study protocol for a randomised controlled trial. *BMC Musculoskeletal Disorders*, *22*(1), 695. https://doi.org/10.1186/s12891-021-04553-6

Hashimoto, T., & Sakuraba, K. (2014, March). Strength training for the intrinsic flexor muscles of the foot: Effects on muscle strength, the foot arch, and dynamic parameters before and after the training. *Journal of Physical Therapy Science*, *26*(3), 373–376. https://doi.org/10.1589/jpts.26.373

Havens, K. L., & Sigward, S. M. (2015, April). Cutting mechanics: Relation to performance and anterior cruciate ligament injury risk. *Medicine & Science in Sports & Exercise*, *47*(4), 818–824. https://doi.org/10.1249/MSS.0000000000000470

Hopkins, W. G., Marshall, S. W., Batterham, A. M., & Hanin, J. (2009, January). Progressive statistics for studies in sports medicine and exercise science. *Medicine and Science in Sports and Exercise*, *41*(1), 3–12. https://doi.org/10.1249/MSS.0b013e31818cb278

Hori, M., Suga, T., Terada, M., Tanaka, T., Kusagawa, Y., Otsuka, M., Nagano, A., & Isaka, T. (2021, June 10). Relationship of the knee extensor strength but not the quadriceps femoris muscularity with sprint performance in sprinters: A reexamination and extension. *BMC Sports Science, Medicine and Rehabilitation*, *13*(1), 67. https://doi.org/10.1186/s13102-021-00293-1

Kessler, S. E., Lichtwark, G. A., Welte, L. K. M., Rainbow, M. J., & Kelly, L. A. (2020, July). Regulation of foot and ankle quasi-stiffness during human hopping across a range of frequencies. *Journal of Biomechanics*, *108*, 109853. https://doi.org/10.1016/j.jbiomech.2020.109853

Kokkonen, J., Bangerter, B., Roundy, E., & Nelson, A. (1988, March 1). Improved performance through digit strength gains. *Research Quarterly for Exercise and Sport*, *59*(1), 57–63. https://doi.org/10.1080/02701367.1988.10605474

Konrad, A., Reiner, M. M., Bernsteiner, D., Glashüttner, C., Thaller, S., & Tilp, M. (2021, March 3). Joint flexibility and isometric strength parameters are not relevant determinants for countermovement jump performance. *International Journal of Environmental Research and Public Health*, *18*(5), 2510. https://doi.org/10.3390/ijerph18052510

Kotsifaki, A., Korakakis, V., Graham-Smith, P., Sideris, V., & Whiteley, R. (2021, March). Vertical and horizontal hop performance: Contributions of the hip, knee, and ankle. *Sports Health: A Multidisciplinary Approach*, *13*(2), 128–135. https://doi.org/10.1177/1941738120976363

Križaj, J., Rauter, S., Vodičar, J., Hadžić, V., & Šimenko, J. (2019, March 4). Predictors of vertical jumping capacity in soccer players. *Isokinetics and Exercise Science*, *27*(1), 9–14. https://doi.org/10.3233/IES-182138

Liew, B. X. W., Sullivan, L., Morris, S., & Netto, K. (2020, June). Mechanical work performed by distal foot-ankle and proximal knee-hip segments during anticipated and unanticipated cutting. *Journal of Biomechanics*, *106*, 109839. https://doi.org/10.1016/j.jbiomech.2020.109839

Linthorne, N. P. (2001, November). Analysis of standing vertical jumps using a force platform. *American Journal of Physics*, *69*(11), 1198–1204. https://doi.org/10.1119/1.1397460

Maniar, N., Schache, A. G., Cole, M. H., & Opar, D. A. (2019, January 3). Lower-limb muscle function during sidestep cutting. *Journal of Biomechanics*, *82*, 186–192. https://doi.org/10.1016/j.jbiomech.2018.10.021

Marshall, B. M., Franklyn-Miller, A. D., King, E. A., Moran, K. A., Strike, S. C., & Falvey, É. C. (2014, October). Biomechanical factors associated with time to complete a change of direction cutting maneuver. *The Journal of Strength & Conditioning Research*, *28*(10), 2845–2851. https://doi.org/10.1519/JSC.0000000000000463

McDonald, K. A., Stearne, S. M., Alderson, J. A., North, I., Pires, N. J., Rubenson, J., & Garcia Aznar, J. M. (2016). The role of arch compression and metatarsophalangeal joint dynamics in modulating plantar fascia strain in running. *Public Library of Science ONE*, *11*(4), e0152602. https://doi.org/10.1371/journal.pone.0152602

McMahon, J. J., Suchomel, T. J., Lake, J. P., & Comfort, P. (2018, August). Understanding the key phases of the countermovement jump force-time curve. *Strength & Conditioning Journal*, *40*(4), 96–106. https://doi.org/10.1519/SSC.0000000000000375

Miller, R., Balshaw, T. G., Massey, G. J., Maeo, S., Lanza, M. B., Johnston, M., Allen, S. J., & Folland, J. P. (2021, April 1). The muscle morphology of elite sprint running. *Medicine and Science in Sports and Exercise*, *53*(4), 804–815. https://doi.org/10.1249/MSS.0000000000002522

Nagahara, R., Mizutani, M., Matsuo, A., Kanehisa, H., & Fukunaga, T. (2018, April). Association of sprint performance with ground reaction forces during acceleration and maximal speed phases in a single sprint. *Journal of Applied Biomechanics*, *34*(2), 104–110. https://doi.org/10.1123/jab.2016-0356

Nagahara, R., Naito, H., Miyashiro, K., Morin, J. B., & Zushi, K. (2014, December). Traditional and ankle-specific vertical jumps as strength-power indicators for maximal sprint acceleration. *The Journal of Sports Medicine and Physical Fitness*, *54*(6), 691–699.

Newman, M. A., Tarpenning, K. M., & Marino, F. E. (2004, November). Relationships between isokinetic knee strength, single-sprint performance, and repeated-sprint ability in football players. *The Journal of Strength & Conditioning Research*, *18*(4), 867–872. https://doi.org/10.1519/00124278-200411000-00032

Pandy, M. G., Lai, A. K. M., Schache, A. G., & Lin, Y. C. (2021, October). How muscles maximize performance in accelerated sprinting. *Scandinavian Journal of Medicine & Science in Sports*, *31*(10), 1882–1896. https://doi.org/10.1111/sms.14021

Péter, A., Hegyi, A., Stenroth, L., Finni, T., & Cronin, N. J. (2015, September 18). EMG and force production of the flexor hallucis longus muscle in isometric plantarflexion and the push-off phase of walking. *Journal of Biomechanics*, *48*(12), 3413–3419. https://doi.org/10.1016/j.jbiomech.2015.05.033

Redmond, A. C., Crosbie, J., & Ouvrier, R. A. (2006, January). Development and validation of a novel rating system for scoring standing foot posture: The foot posture index. *Clinical Biomechanics*, *21*(1), 89–98. https://doi.org/10.1016/j.clinbiomech.2005.08.002

Roma, E., Michel, A., Tourillon, R., Millet, G. Y., & Morin, J. B. (2024, June 1). Reliability and measurement error of a maximal voluntary toe plantarflexion measurement process. *The Foot*, *59*, 102095. https://doi.org/10.1016/j.foot.2024.102095

Saeki, J., Iwanuma, S., & Torii, S. (2021, January 12). Force generation on the hallux is more affected by the ankle joint angle than the lesser toes: An in vivo human study. *Biology (Basel)*, *10*(1), 48. https://doi.org/10.3390/biology10010048

Schache, A. G., Brown, N. A. T., & Pandy, M. G. (2015, August). Modulation of work and power by the human lower-limb joints with increasing steady-state locomotion speed. *The Journal of Experimental Biology*, *218*(Pt 15), 2472–2481. https://doi.org/10.1242/jeb.119156

Schache, A. G., Lai, A. K. M., Brown, N. A. T., Crossley, K. M., & Pandy, M. G. (2019, November 15). Lower-limb joint mechanics during maximum acceleration sprinting. *The Journal of Experimental Biology*, *222*(22), jeb209460. https://doi.org/10.1242/jeb.209460

Schneiders, A. G., Sullivan, S. J., O'Malley, K. J., Clarke, S. V., Knappstein, S. A., & Taylor, L. J. (2010, September). A valid and reliable clinical determination of footedness. *PM&R*, *2*(9), 835–841. https://doi.org/10.1016/j.pmrj.2010.06.004

Smith, G., Lake, M., & Lees, A. (2014, April). Metatarsophalangeal joint function during sprinting: A comparison of barefoot and sprint spike shod foot conditions. *Journal of Applied Biomechanics*, *30*(2), 206–212. https://doi.org/10.1123/jab.2013-0072

Smith, R., Lichtwark, G., Farris, D., & Kelly, L. (2022, July). Examining the intrinsic foot muscles' capacity to modulate plantar flexor gearing and ankle joint contributions to propulsion in vertical jumping. *Journal of Sport and Health Science*, *12*(5), S639–647. https://doi.org/10.1016/j.jshs.2022.07.002

Smith, R. E., Lichtwark, G. A., & Kelly, L. A. (2021). The energetic function of the human foot and its muscles during accelerations and decelerations. *Journal of Experimental Biology*, *224*(13), jeb242263. https://doi.org/10.1242/jeb.242263

Spiteri, T., Cochrane, J. L., Hart, N. H., Haff, G. G., & Nimphius, S. (2013, November). Effect of strength on plant foot kinetics and kinematics during a change of direction task. *European Journal of Sport Science*, *13*(6), 646–652. https://doi.org/10.1080/17461391.2013.774053

Stefanyshyn, D. J., & Nigg, B. M. (1998, August). Dynamic angular stiffness of the ankle joint during running and sprinting. *Journal of Applied Biomechanics*, *14*(3), 292–299. https://doi.org/10.1123/jab.14.3.292

Tanaka, T., Suga, T., Imai, Y., Ueno, H., Misaki, J., Miyake, Y., Otsuka, M., Nagano, A., & Isaka, T. (2019, April 21). Characteristics of lower leg and foot muscle thicknesses in sprinters: Does greater foot muscles contribute to sprint performance? *European Journal of Sport Science*, *19*(4), 442–450. https://doi.org/10.1080/17461391.2018.1534991

Tourillon, R., Six, A., Bothorel, H., & Fourchet, F. (2023, March 24). Are foot posture and morphological deformation associated with ankle plantar flexion isokinetic strength and vertical drop jump kinetics? A principal component analysis. *Sports Biomechanics*, 1–15. https://doi.org/10.1080/14763141.2023.2191868

Turner, A. N., Read, P., Maestroni, L., Chavda, S., Yao, X., Papadopoulos, K., Virgile, A., Spiegelhalter, A., & Bishop, C. (2022, August). Reverse engineering in strength and conditioning: Applications to agility training. *Strength & Conditioning Journal*, *44*(4), 85. https://doi.org/10.1519/SSC.0000000000000681

Unger, C. L., & Wooden, M. J. (2000, November). Effect of foot intrinsic muscle strength training on jump performance. *The Journal of Strength & Conditioning Research*, *14*(4), 373–378. https://doi.org/10.1519/00124278-200011000-00001

Vanezis, A., & Lees, A. (2005, November 15). A biomechanical analysis of good and poor performers of the vertical jump. *Ergonomics*, *48*(11–14), 1594–1603. https://doi.org/10.1080/00140130500101262

Wdowski, M. M., Rosicka, K., & Hill, M. (2023, January). Influence of lower-limb muscular and tendon mechanical properties and strength on countermovement jump performance. *The Journal of Sports Medicine and Physical Fitness*, *63*(1), 16–22. https://doi.org/10.23736/S0022-4707.22.13567-X

Wu, C. X., Liu, H. B., Zhao, Z. N., Wang, Y. B., & Luan, Z. L. (2022, December 1). Effects of movement direction and limb dominance on ankle muscular force in sidestep cutting. *Medical Engineering & Physics*, *110*, 103914. https://doi.org/10.1016/j.medengphy.2022.103914

Yamauchi, J., & Koyama, K. (2020, May). Importance of toe flexor strength in vertical jump performance. *Journal of Biomechanics*, *104*, 109719. https://doi.org/10.1016/j.jbiomech.2020.109719

Yuasa, Y., Kurihara, T., & Isaka, T. (2018, October 15). Relationship between toe muscular strength and the ability to change direction in athletes. *Journal of Human Kinetics*, *64*(1), 47–55. https://doi.org/10.1515/hukin-2017-0183